

Question1

Q1.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 1</title>
</head>
<body>

  <h1>Question 1 - Basic Array Methods</h1>

  <script src="q1.js"></script>

</body>
</html>
```

Q1.js:

```
let fruits = ["peach", "cherry", "blackberry", "lyche", "Guava"];

console.log("Number of fruits:", fruits.length);

console.log("Array as string:", fruits.toString());

console.log("First fruit:", fruits.at(0));

console.log("Third fruit:", fruits.at(2));

console.log("Fruits with separator:", fruits.join(" - "));
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports
Number of fruits: 5
Array as string: peach,cherry,blackberry,lyche,Guava
First fruit: peach
Third fruit: blackberry
Fruits with separator: peach - cherry - blackberry - lyche - Guava
```

Question2

Q2.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 2</title>
```

```
</head>
<body>

  <h1>Question 2 - Adding and Removing Students</h1>

  <script src="q2.js"></script>

</body>
</html>
```

Q2.js:

```
// Question 2: Adding and Removing Array Elements

let students = ["Ali", "Ahmed", "Fatima", "Sara", "Hassan"];

console.log("Original array:", students);

// push()
students.push("Ayesha");
console.log("After push:", students);

// pop()
let removedStudent = students.pop();
console.log("Removed by pop:", removedStudent);
console.log("After pop:", students);

// unshift()
students.unshift("Usman");
console.log("After unshift:", students);

// shift()
let firstStudent = students.shift();
console.log("Removed by shift:", firstStudent);
console.log("After shift:", students);

// length
console.log("Final number of students:", students.length);
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports  Filter (e.g
> Original array: (5) ['Ali', 'Ahmed', 'Fatima', 'Sara', 'Hassan']
> After push: (6) ['Ali', 'Ahmed', 'Fatima', 'Sara', 'Hassan', 'Ayesha']
  Removed by pop: Ayesha
> After pop: (5) ['Ali', 'Ahmed', 'Fatima', 'Sara', 'Hassan']
> After unshift: (6) ['Usman', 'Ali', 'Ahmed', 'Fatima', 'Sara', 'Hassan']
  Removed by shift: Usman
> After shift: (5) ['Ali', 'Ahmed', 'Fatima', 'Sara', 'Hassan']
  Final number of students: 5
```

Question 3

Q3.html

```
<!DOCTYPE html>

<html>
<head>
  <title>Question 3</title>
</head>
<body>

  <h1>Question 3 - Array Modification</h1>

  <script src="q3.js"></script>

</body>
</html>
```

Q3.js:

```
// Question 3: concat(), slice(), splice() and delete

let fruits = ["Apple", "Banana", "Mango", "Orange"];
let vegetables = ["Potato", "Tomato", "Carrot"];

// concat()
let combined = fruits.concat(vegetables);
console.log("Combined array:", combined);

// slice()
let selected = fruits.slice(1, 3);
console.log("Slice array:", selected);

// splice() - remove one element
fruits.splice(1, 1);
console.log("After removing with splice:", fruits);

// splice() - add an element
fruits.splice(1, 0, "Banana");
console.log("After adding with splice:", fruits);

// delete
delete fruits[2];

console.log("After delete:", fruits);
console.log("Value at deleted position:", fruits[2]);
console.log("Length after delete:", fruits.length);
```

Output:

```
Problems Output Debug Console Terminal Ports Filter (e.g. text, !exclude, \esca...
> Combined array: (7) ['Apple', 'Banana', 'Mango', 'Orange', 'Potato', 'Tomato', 'Carrot']
> Slice array: (2) ['Banana', 'Mango']
> After removing with splice: (3) ['Apple', 'Mango', 'Orange']
> After adding with splice: (4) ['Apple', 'Banana', 'Mango', 'Orange']
> After delete: (4) ['Apple', 'Banana', ..., 'Orange']
  Value at deleted position: undefined
  Length after delete: 4
>
```

Question4

Q4.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 4</title>
</head>
<body>

  <h1>Question 4 - Array.isArray() and Arrow Functions</h1>

  <script src="q4.js"></script>

</body>
</html>
```

Q4.js:

```
// Question 4: Array.isArray() and Arrow Functions

let colors = ["Red", "Blue", "Green"];
let name = "Fatima";
let age = 22;

// Array.isArray()
console.log("Is colors an array?", Array.isArray(colors));
console.log("Is name an array?", Array.isArray(name));
console.log("Is age an array?", Array.isArray(age));

// Arrow function with an array
let showArray = (array) => {
  console.log("Array:", array);
};

// Calling the arrow function
```

```
showArray(colors);

// Another arrow function
let showValue = (value) => {
  console.log("Value:", value);
};

// Calling the function
showValue("Hello JavaScript");
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports
Is colors an array? true
Is name an array? false
Is age an array? false
> Array: (3) ['Red', 'Blue', 'Green']
Value: Hello JavaScript
```

Question 5

Q5.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 5</title>
</head>
<body>

  <h1>Question 5 - Product List Manager</h1>

  <script src="q5.js"></script>

</body>
</html>
```

Q5.js:

```
// Question 5: Product List Manager

let products = [
  "Laptop",
  "Mouse",
  "Keyboard",
```

```
    "Monitor",
    "Headphones"
];

// length
console.log("Number of products:", products.length);

// at()
console.log("First product:", products.at(0));
console.log("Last product:", products.at(-1));

// push()
products.push("Printer");
console.log("After push:", products);

// unshift()
products.unshift("Webcam");
console.log("After unshift:", products);

// pop()
products.pop();
console.log("After pop:", products);

// shift()
products.shift();
console.log("After shift:", products);

// Second product array
let moreProducts = ["USB", "Speaker"];

// concat()
let allProducts = products.concat(moreProducts);
console.log("After concat:", allProducts);

// slice()
let smallList = allProducts.slice(0, 3);
console.log("Smaller list:", smallList);

// splice()
allProducts.splice(1, 1, "Wireless Mouse");
console.log("After splice:", allProducts);

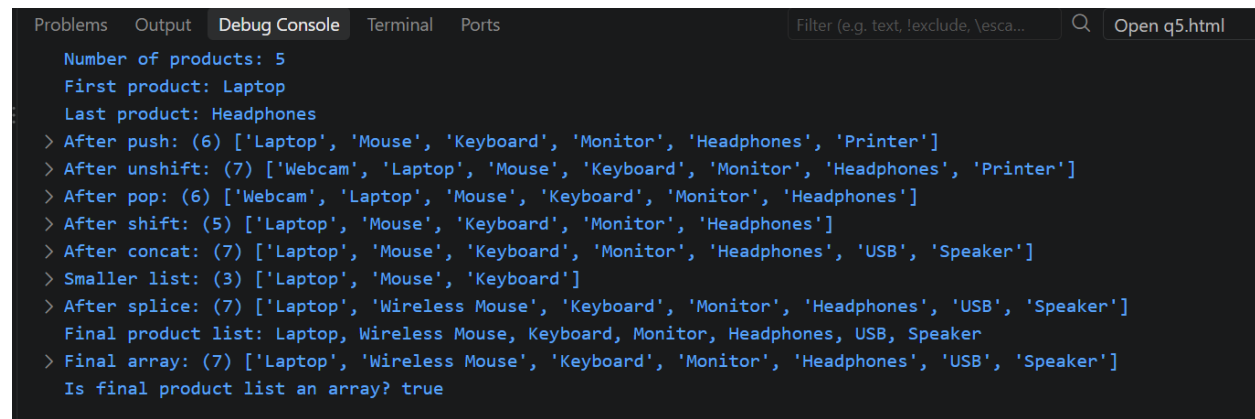
// join()
console.log("Final product list:", allProducts.join(", "));
```

```
// Arrow function
let showProducts = (array) => {
  console.log("Final array:", array);
};

showProducts(allProducts);

// Array.isArray()
console.log("Is final product list an array?", Array.isArray(allProducts));
```

Output:



```
Problems Output Debug Console Terminal Ports
Filter (e.g. text, !exclude, \esca... Q Open q5.html

Number of products: 5
First product: Laptop
Last product: Headphones
> After push: (6) ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones', 'Printer']
> After unshift: (7) ['Webcam', 'Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones', 'Printer']
> After pop: (6) ['Webcam', 'Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones']
> After shift: (5) ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones']
> After concat: (7) ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones', 'USB', 'Speaker']
> Smaller list: (3) ['Laptop', 'Mouse', 'Keyboard']
> After splice: (7) ['Laptop', 'Wireless Mouse', 'Keyboard', 'Monitor', 'Headphones', 'USB', 'Speaker']
Final product list: Laptop, Wireless Mouse, Keyboard, Monitor, Headphones, USB, Speaker
> Final array: (7) ['Laptop', 'Wireless Mouse', 'Keyboard', 'Monitor', 'Headphones', 'USB', 'Speaker']
Is final product list an array? true
```